Passport-to-SmartLock Gate

A SaaS for Dubai holiday-home hosts: guest ID registration and door access in one automated flow

What this guide covers: the concept in plain English, what the Dubai rules actually say (verified in Oct 2026), the competition, guest-journey flowchart, system architecture, a 10–12 week build plan, pricing maths, legal/privacy risks, and your first-10-customers plan.

Honest headline: the problem is real, but this is **not an empty "blue ocean"**. Chekin, ChargeAutomation and others already automate Dubai guest registration, and Dubai's DET has launched its own contactless check-in capability. Your opportunity is a narrower wedge, explained in Section 3.

1. The concept in plain English

The problem. Dubai requires licensed holiday homes (Airbnb-style apartments) to register every guest with the government tourism system, using a passport or Emirates ID. Hosts and property managers today chase tourists on WhatsApp for passport photos, type the details into the government portal by hand, and only then send the door code. Late arrivals at 2 AM make this painful, and mistakes risk penalties.

Your product. Put the door PIN *behind* the registration step. The guest gets a WhatsApp link, scans their passport and selfie in about a minute, your system checks the data, files the registration, and only then creates a time-limited PIN on the smart lock. No scan means no PIN, so non-compliant stays cannot happen by accident.

30-second pitch. “I make sure every guest is registered with Dubai's tourism department before they can open the door. The host does nothing: the guest scans a passport on WhatsApp, and the smart lock unlocks itself once registration succeeds.”

Who pays and why

Customer	Pain today	Why they would pay
Individual owners (up to 8 units)	Manual registration, midnight WhatsApp chasing, fear of penalties	Saves hours per month; peace of mind
Small holiday-home operators (5–100 units)	Staff time per guest, errors, missed deadlines	Cheaper than hiring another coordinator
Property managers with smart locks already	Lock system and compliance system are separate tools	One flow instead of two

Why “gate the PIN” is the clever bit

- It turns compliance from a chore into a **built-in step of getting in**, so hosts feel the value immediately.
- It gives you a proof trail: timestamped registration ID plus lock-open log for every stay.
- It is hard for a generic global tool to copy exactly because it needs Dubai-specific rules plus lock integration.

2. What the Dubai rules actually say (verified)

The source playbook mentioned “DTCM”, “tourism police” and a “AED 10,000 fine”. Here is what I could confirm from official and industry pages, and what I could not.

Topic	What sources say	Confidence
Regulator	Dubai Department of Economy and Tourism (DET), formerly DTCM. Holiday homes are managed through DET's Holiday Homes system (HH 2.0).	High (DET site)
Licensing	Every holiday-home unit needs registration. Owners can register up to 8 units; registration fee about AED 1,500 + AED 20 in knowledge/innovation fees.	High (DET site)
Guest ID	A passport or Emirates ID photo for every guest must be uploaded to the system.	High
Deadline	Several operator/industry guides say guests must be registered within **3 hours of check-in**. One guide says 24 hours (to the federal ICA). Not the same as “before they step inside”.	Medium: confirm with DET (call 600545555)
Tourism Dirham	Per-room-per-night fee collected from guests and remitted to DET (AED 10–15 depending on category).	High (DET site)
Fine amounts	Guides mention penalties and licence suspension; one cites up to AED 100,000 for violations like unlicensed operation. The playbook's “AED 10,000 for missed passport upload” was **not confirmed**.	Low: do not quote a number in marketing until verified
DET contactless check-in	In Dec 2025 Dubai announced a citywide one-time contactless check-in capability built by DET and available via authorised technology providers.	High (WAM / Dubai Media Office)

Action item #1: before writing code, call DET (600545555) or ask a licensed operator: how does a third-party software submit guests to HH 2.0 (official API, authorised-provider programme, or portal-only)? The answer decides your architecture and timeline.

3. Competition and your wedge

Because the idea is public-facing compliance, others have noticed it too:

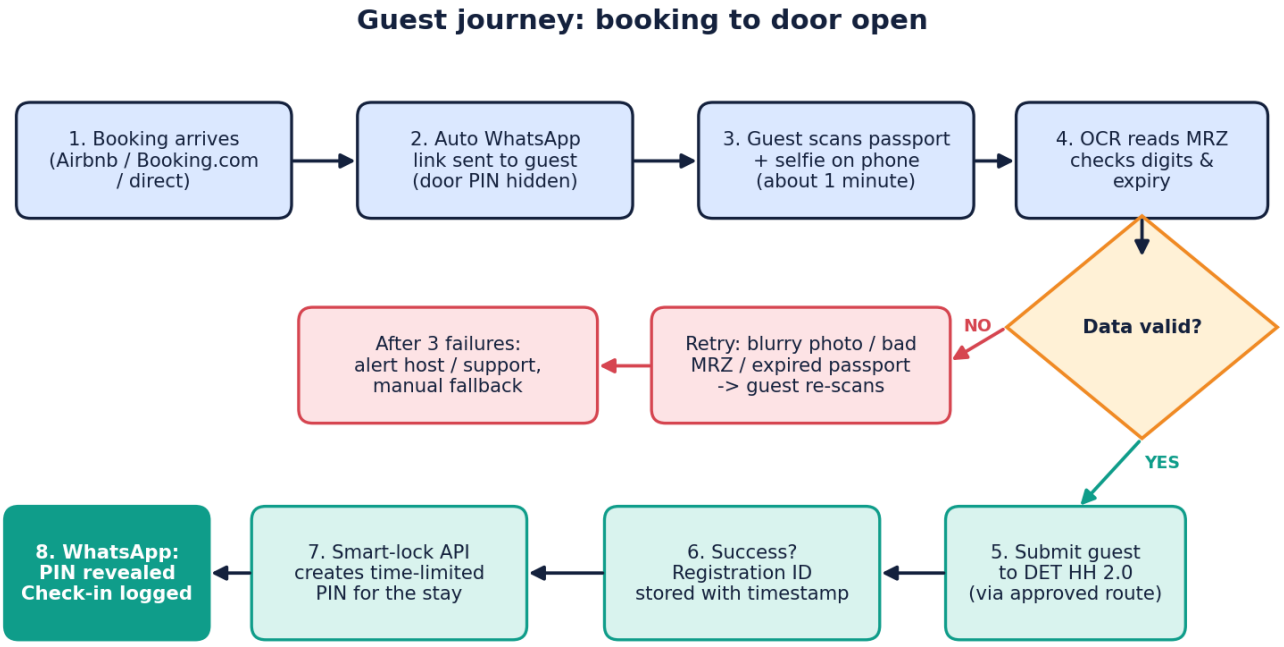
Player	What they do in Dubai	Gap you could use
Chekin	Automated DET registration, ID capture, 50+ PMS integrations, partner with TTLock	Generic, priced for professional PMS users
ChargeAutomation	Integrates with DTCM HH 2.0 and EMAAR reporting; guest check-in and payments	Broad product, not lock-first
Proofsnap-type tools	Evidence capture for hosts	Not an access-control product
Operator-run smart locks (e.g. Deluxe)	Own lock + check-in for their own managed units	Not available to independent owners
DET + authorised providers	Official one-time contactless check-in	Your tool could be an *integration* of this, not a rival

Where a solo developer can still win

- **Tiny hosts and small operators** (1–30 units) who find enterprise PMS-based tools too heavy or expensive. Price at about AED 39 per apartment per month.
- **Lock-first workflow:** sell it as “smart-lock check-in that is automatically compliant”, with registration as the built-in step rather than a separate product.
- **Indian-run operators in Dubai:** many are reachable in English/Hindi on WhatsApp and Zoom from India.
- **Use DET's official contactless check-in** where available instead of inventing your own identity flow: faster approval and more trust.

Rule of thumb: if after 10 conversations no small host says “I would pay AED 39 for this today”, pivot to a different idea from your list rather than building. Validate first.

4. How it works: guest journey

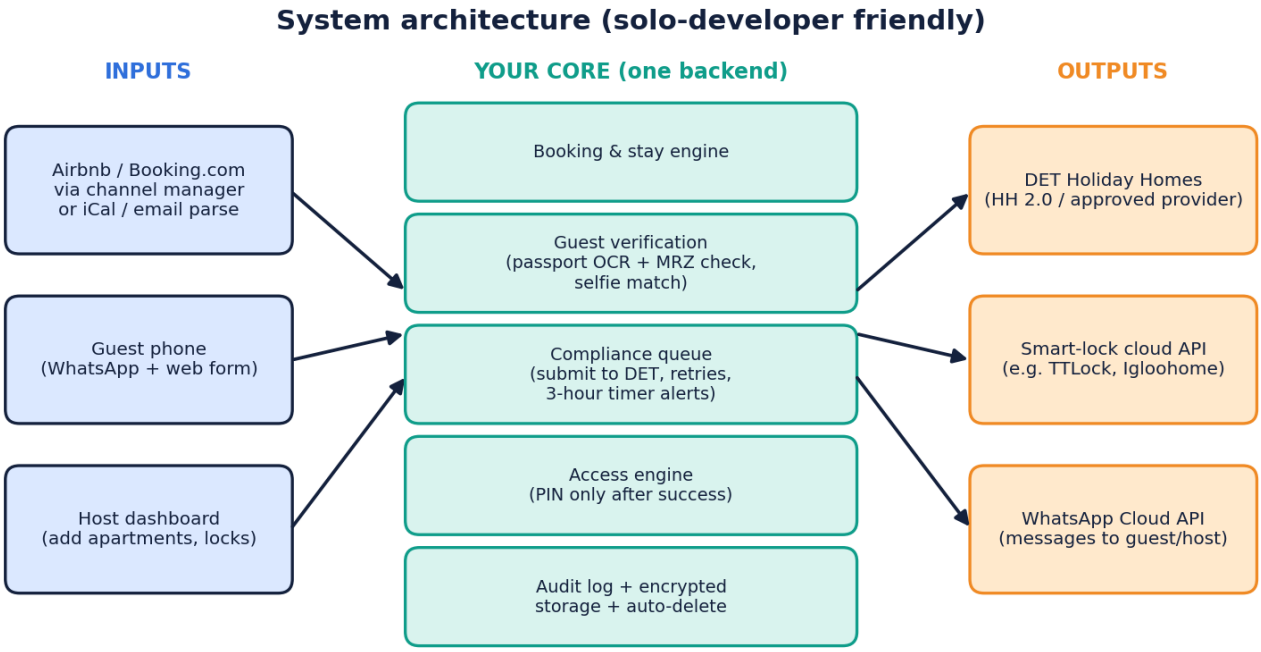


Key rule: the PIN is only generated AFTER the registration step succeeds.

Design principles

- **Fail safe:** the host always has a manual override and gets alerted if the guest is stuck.
- **Timers:** track the registration deadline from check-in and alert the host before it expires.
- **Time-limited PINs:** create the lock code only for the stay window and delete it after checkout.
- **Group bookings:** collect ID for every adult guest, not just the person who booked.

5. System architecture



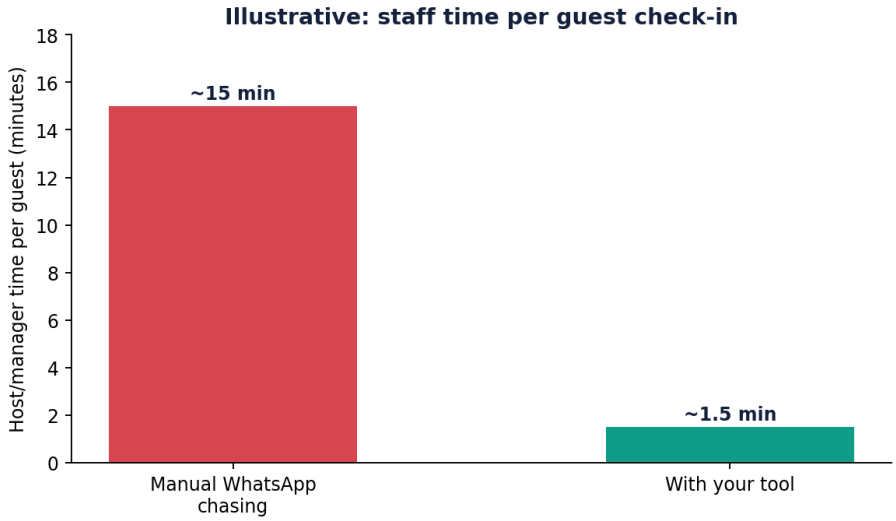
Suggested stack: Python/FastAPI or Node, Postgres, Redis queue, S3-style encrypted storage, hosted in UAE/GCC region.

What each part does

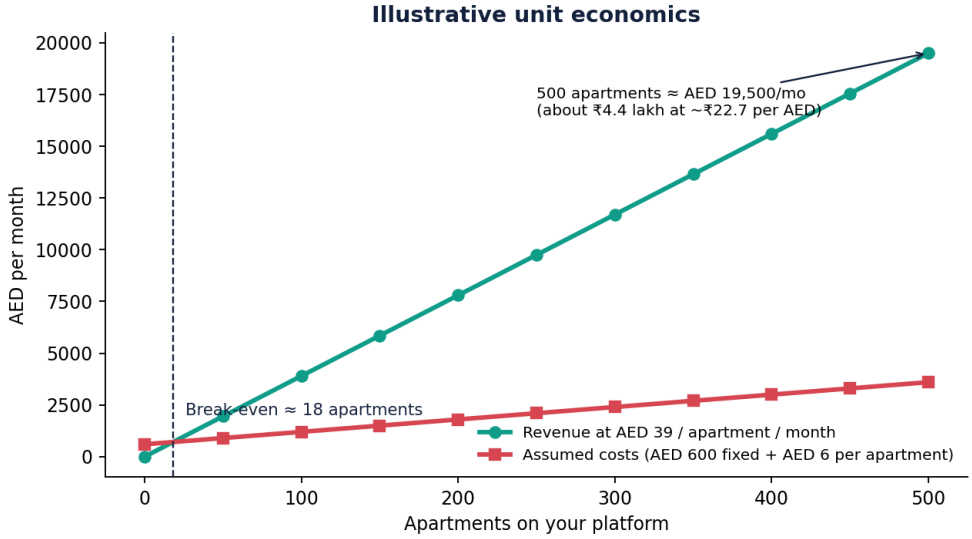
Component	Purpose	Practical tools
Booking engine	Know who arrives when, in which unit	Channel-manager API, iCal feeds, or parse confirmation emails to start
Guest verification	Read passport MRZ, validate check digits and expiry, optional selfie match	Open-source MRZ/OCR libraries, or an identity-verification API (paid, faster)
Compliance queue	Submit to DET, retry on failure, store registration ID, run deadline alerts	Redis/BullMQ or Celery; worker with retries
Access engine	Create/delete time-limited PINs after success	Smart-lock cloud APIs (e.g. TTLock Open Platform, Igloohome) or an aggregator
Messaging	Send links, PINs and alerts	WhatsApp Business Cloud API; SMS fallback
Security layer	Encrypt ID images, auto-delete, audit log	Encrypted object storage in UAE/GCC region, row-level access

6. Value and unit economics

The graphs below use **assumed numbers for illustration**. Replace them with real figures from your pilot.

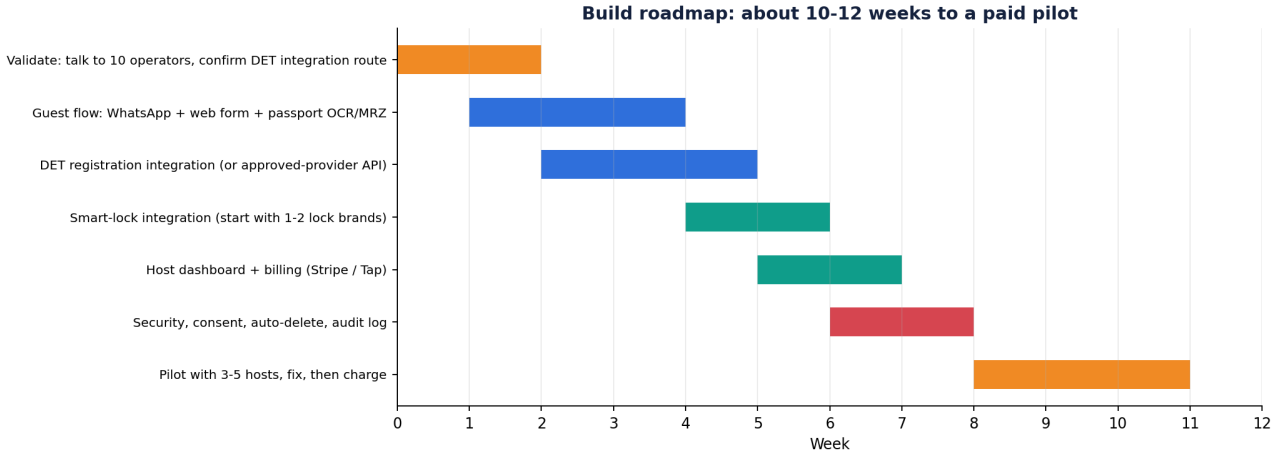


Assumed figures for illustration only. Measure real numbers with 3-5 pilot hosts.



Pricing options to test: AED 39 per apartment per month (simple), or AED 99 flat for up to 3 units plus AED 29 per extra unit. Offer a free first month for pilot hosts, then charge. Currency note: AED to INR is roughly ₹22–23; check the live rate.

7. How to build it: step by step



Phase 0: Validate (weeks 0–2)

- Talk to 10 holiday-home owners/operators (Facebook groups, Dubai property-manager WhatsApp groups, LinkedIn).
- Ask: how do you register guests today, how long does it take, what tools do you use, what would you pay?
- Call DET to confirm how third-party software can submit guests. Ask about authorised technology providers.

Phase 1: MVP (weeks 1–7)

- **Guest web form + WhatsApp link:** mobile web page that opens the camera, captures the passport, runs MRZ validation.
- **Registration step:** start with the route DET allows. If portal-only, an assisted workflow (host one-click approve) is a valid MVP while you apply for official access.
- **Lock integration:** pick ONE brand most Dubai hosts use (check TTLock first), generate and delete time-bound passcodes through its cloud API.
- **Host dashboard:** add apartments, link locks, see each stay's status (waiting, registered, PIN sent, checked in).
- **Booking input:** begin with manual entry plus iCal/email parsing; add channel-manager APIs later.

Phase 2: Pilot and charge (weeks 8–12)

- Run with 3–5 friendly hosts, shadow every check-in, log failures (blurry photos, group bookings, lock offline).
- Add billing (Stripe or Tap Payments), email/WhatsApp support, and a data-deletion schedule.
- Start charging after the first month. Aim for 10 paying units before adding features.

8. Legal, privacy and technical risks

Risk	Why it matters	Mitigation
Passport data is highly sensitive	UAE personal data protection law and DET data-handling rules apply; a leak would end the business	Encrypt at rest, store in UAE/GCC region, minimum retention, delete after the required period, strict access logs. Get a UAE lawyer to review.
Unclear integration route to DET	If there is no open API, scraping the portal may breach terms	Use only official routes or authorised-provider programme; start assisted/manual if needed
Smart-lock failures	Guest locked out at 2 AM = angry host	Offline Bluetooth fallback, 24/7 host alert, backup key box
Wrong deadline assumption	Sources differ (3 hours vs 24 hours)	Confirm with DET; build the timer as a setting
Fraudulent IDs	You may be blamed if bad data is submitted	Check digits, expiry, selfie match; keep host responsible by contract
Competition copies you	Chekin/ChargeAutomation could add lock-gating	Move fast, own the small-host niche, bundle support in English/Hindi
Selling from India to Dubai	Need local invoicing, VAT (5%) questions	Use a payment provider that supports AED; ask an accountant about invoicing entity

Disclaimer: this guide is research and planning help, not legal advice. Verify regulations directly with DET and a UAE-licensed lawyer before launch.

9. First 10 customers plan

- **Week 1–2:** join Dubai holiday-home owner and property-manager groups; offer a free “compliance and check-in audit” in exchange for a 15-minute call.
- **Week 3–4:** demo with a simulated guest scan on your phone. The 60-second demo sells itself: scan passport, door code appears.
- **Week 5–8:** sign 3–5 pilot hosts free for one month, collect testimonials and screenshots.
- **Week 9–12:** convert pilots to paid, ask each for two referrals, and approach small operators with 10–30 units.
- **Channels that fit a solo founder in India:** LinkedIn outreach, WhatsApp groups, Indian CA and property-management networks in Dubai, Facebook groups for Dubai hosts.

Success metrics for the first 90 days

Metric	Target
Hosts interviewed	10+
Pilot hosts live	3–5
Paying apartments	10–30
Guest completion rate (scan to PIN)	above 90%
Median time from link to PIN	under 3 minutes

Your next 5 actions

- Call DET (600545555) and ask how software integrates with Holiday Homes / the contactless check-in providers.
- Interview 10 hosts or operators this week.
- Open a TTLock developer account and test creating a time-limited passcode.
- Prototype the passport-scan web page (MRZ read + validation) as a weekend project.
- Decide: continue or switch ideas based on whether hosts say they would pay.